



# Gayathri Rangu

Roll No.:216101003

Ph.D - Computer Science and Engineering

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Github | LinkedIn | Google Scholar

## EDUCATION

| Degree/Certificate | Institute/Board                                     | CGPA/Percentage | Year      |
|--------------------|-----------------------------------------------------|-----------------|-----------|
| Ph.D.              | Indian Institute of Technology Guwahati             | 7.83            | 2021-2026 |
| B.Tech             | Jawaharlal Nehru Technological University Hyderabad | 8.58            | 2017-2021 |
| Intermediate       | Telangana State Board of Intermediate Education     | 96.4%           | 2017      |
| Secondary          | Saint Mary's High School (SSC)                      | 9.3             | 2015      |

## RESEARCH

Researcher specializing in **Federated Reinforcement Learning (FRL)**, **multi-robot systems**, and **embodied AI**. My work focuses on developing scalable learning frameworks for distributed robotic systems, including **selective aggregation**, **prioritized experience replay**, **decentralized swarm learning**, **unsupervised skill discovery**, and **sim-to-real transfer**. Experienced in implementing **Deep Reinforcement Learning (DRL)** algorithms and robotic learning pipelines across simulators and real robotic platforms, with additional interests in **LLM-assisted robotics** and **autonomous AI systems**.

## TEACHING ASSISTANTSHIP

- **Head TA - CS321 Smart Systems Lab** Spring 2026, Spring 2023
- **Head TA - CS1102 Programming in C** Fall 2025
- **Head TA - CS561 Artificial Intelligence** Spring 2025
- **Head TA - CS666 Mobile Robotics Lab** Fall 2024, Fall 2023
- **Head TA - CS101 Introduction to Computing** Spring 2022
- **TA - CS565 Intelligent Systems and Interfaces** Fall 2021

## INDUSTRY EXPERIENCE

- **Qualcomm** Feb. 2024 – Aug. 2024  
Research Intern  
Bangalore, India
  - Developed **AI/ML pipelines** on **Qualcomm QRB5-powered robotic platforms** for edge-based robotic applications.
  - Built an **LLM-assisted ticket retrieval system** for identifying semantically similar entries from internal Lessons Learnt (LL) JIRA repositories.
  - Contributed AI models and deployment workflows to the **Qualcomm AI Hub** platform.
- **CtrlS Datacenters Ltd.** Dec. 2020 – Jun. 2021  
Network Engineering Intern  
Hyderabad, India
  - Worked on network monitoring, ticket resolution, and network simulation workflows using **Cisco Packet Tracer**.

## RESEARCH PUBLICATIONS

- **Last Skills Standing: Adaptive Skill Discovery in Unsupervised Reinforcement Learning** SCITEPRESS  
International Conference on Agents and Artificial Intelligence (ICAART 2026), Marbella, Spain
- **Real-Time Information Synthesis for Heterogeneous Sources Using Adaptive LLM-Based Summarization** Submitted  
2026 IEEE Region 10 Conference (TENCON), Bali, Indonesia
- **On Enhancing Sim-to-Real Transfer in Robots via Federated Reinforcement Learning** Submitted  
2026 IEEE Region 10 Conference (TENCON), Bali, Indonesia
- **Federated Reinforcement Learning for Robots: Augmenting Selective Aggregation with Prioritized Replay** Under Review  
Artificial Intelligence Journal (AIJ)
- **Smarter Robots, Fewer Queries: A Knowledge-Driven Approach to Reduce LLM Dependency** Accepted  
Advances in Robotics (AIR 2025), IIT Jodhpur
- **On using Mobile Agents for Programming Network of Devices** Accepted  
Research and Industrial Conclave (RIC 2025), IIT Guwahati
- **An Enhanced Path Planning strategy for Mobile robots using Aerial Information from Drones** Submitted  
12th European Conference on Mobile Robots (ECMR 2025), University of Padua, Italy
- **Immuno-inspired Selective Aggregation for Decentralized Federated Deep Reinforcement Learning** ACM  
Genetics and Evolutionary Computation Conference (GECCO) 2024, Melbourne, Australia
- **On Mobile-Agent-based Swarm Reinforcement Learning in a Heterogeneous Robotic Network** ACM  
Advances in Robotics (AIR 2023), IIT Ropar
- **A Hybrid Federated Reinforcement Learning Approach for Networked Robots** Springer  
International Conference on Science, Technology and Engineering (ICSTE 2023)

## PROJECTS

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- **Aerial Guardian: Drone-based Multi-Object Person Tracking** May 2026 – Jun. 2026  
*Computer Vision Assignment* github
  - Developed an aerial person detection and multi-object tracking pipeline using the VisDrone MOT dataset.
  - Fine-tuned **YOLO26m** for small-object aerial detection and integrated **ByteTrack** and **BoT-SORT** with Global Motion Compensation (GMC).
  - Achieved **40+ FPS** real-time performance on a Tesla P100 GPU, improving tracking performance from **24.2%** to **28.3% MOTA** and **38.8% IDF1**.
- **Real-time Information Synthesis Agent (RIS Agent)** Jan. 2025 – Apr. 2026  
*MadScientist* website
  - Developed an autonomous **LLM-based information synthesis pipeline** for real-time ingestion, summarization, and aggregation of technical content from papers, articles, and repositories.
  - Built adaptive summarization workflows using **Llama 3**, **Qwen**, and **GPT-OSS**, incorporating chunking, embeddings, and semantic clustering for long-document processing.
  - Led collaboration with a **6-member research team** and evaluated system performance using **ROUGE** and semantic similarity metrics against state-of-the-art baselines.
- **KERNEL: Knowledge-Embedded Robot Navigation using LLMs** Dec. 2024 – Jul. 2025  
*Research Collaboration*
  - Developed an **LLM-assisted robotic navigation framework** integrating persistent knowledge bases for object localization, exploration, and multi-robot coordination in simulated environments.
  - Implemented knowledge reuse and embedding-based retrieval mechanisms to reduce redundant LLM queries during sequential and multi-object robotic tasks.
  - Evaluated the framework across single-robot and multi-robot scenarios, achieving up to **45% reduction in LLM calls** and improved task completion efficiency.
- **Deep Reinforcement Learning Implementations** Jul. 2023 – Dec. 2023  
*Self Project* github
  - Implemented Deep RL algorithms including **DQN**, **Double DQN**, **PER**, **DDPG**, and **SAC** from scratch using **PyTorch**.
  - Evaluated learning performance across robotic and control environments using **Webots**, **Gymnasium**, and custom simulation pipelines.
  - Experimented with prioritized replay mechanisms, distributed learning setups, and robotic navigation tasks.
- **Digit Recognition through Speech using HMM** Sep. 2021 – Nov. 2021  
*Course Project – IIT Guwahati* github
  - Built a speech-based digit recognition system from scratch in **C++** using **Hidden Markov Models (HMMs)** and custom-recorded datasets.

## TECHNICAL SKILLS

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- **Programming:** Python | C++ | C | Java
- **AI/ML:** PyTorch | TensorFlow | Reinforcement Learning | Federated Learning | LLM Systems
- **Robotics:** *Webots*<sup>®</sup> | MuJoCo | Gymnasium | SLAM | Multi-Agent Systems | Robot Learning
- **Systems & Tools:** Linux | Git | Socket Programming | MQTT
- **Hardware Platforms:** Qualcomm RB5 | mBot Mega | FireBird V | LEGO Mindstorms | Raspberry Pi | Arduino

## LEADERSHIP & ACADEMIC SERVICE

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- **Chairperson, IEEE Student Branch, Guwahati Subsection:** Led technical initiatives and student activities for the 2024–2025 term.
- **General Secretary, IEEE Student Branch, Guwahati Subsection:** Coordinated organizational and outreach activities during the 2023–2024 term.
- **External Reviewer, ACM/IEEE HRI 2026:** Reviewed research submissions in robotics and Human-Robot Interaction domains.
- **Organizing Team, VLSID 2025:** Contributed to organizing activities for the VLSID 2025 conference held in Bangalore.
- **Community Outreach:** Volunteered in educational mentoring and coordination activities through NGO-led student programs.

## ACHIEVEMENTS & ADDITIONAL ACTIVITIES

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- **IEEE Recognition:** Invited as **Guest of Honor** at an IEEE technical event for contributions towards IEEE activities at IIT Guwahati.
  - **Travel Grant:** Received a **USD 450 travel grant** to attend **GECCO 2024**.
  - **GATE 2021:** Secured **96.9 percentile** in GATE CSE 2021.
  - **Gold Medalist:** Department topper at GNIT Hyderabad with the **highest CGPA** in the college.
  - **Best Paper Presentation Award:** Secured **1st Prize** at **SPOORTHY 2019**, a national-level technical symposium.
  - **INDOML 2025 Graduate Forum:** Selected to attend the Graduate Forum at **INDOML 2025**.
  - **Endurance Sports:** Long-distance runner with participation in multiple **half-marathons (21 km)** and Himalayan trekking expeditions.
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